

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Crop Recommendation	Predicts the most suitable crop based on soil nutrients and environmental conditions using machine learning.	ML Prediction Engine	Done	High
2	Data Preprocessing	Cleans the dataset, handles missing values and outliers, and performs feature scaling for accurate model training.	Data Processing	Done	High
3	Machine Learning Model Training	Trains and compares multiple ML algorithms (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) and selects the best model.	ML Training Module	Done	High
4	Web-Based User Interface	Provides an interactive form for users to enter soil and weather parameters and view crop recommendations.	Frontend (HTML/CSS/JS)	Done	Medium
5	Flask Backend API	Receives user inputs, validates data, loads the trained model, and returns crop predictions.	Backend (Flask)	Done	High
6	Input Validation & Error Handling	Validates user inputs and prevents invalid values from reaching the prediction model.	Backend Validation	Done	Medium
7	Model Serialization	Saves the trained model and scaler using Joblib for efficient reuse without retraining.	Model Management	Done	Medium
8	Future Weather API Integration	Automatic retrieval of real-time weather data to improve recommendation accuracy.	API Integration	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Crop Prediction Form, Result Display
2	Backend / Logic	3	Flask API, Input Validation, Crop Recommendation Logic
3	Database / Storage	2	Crop Dataset (CSV), Joblib Model & Scaler Storage
4	API / Integration	1	Flask REST API (<i>Future: OpenWeatherMap API</i>)
5	Security / Authentication	1	Input Validation and Error Handling (<i>User Authentication planned for future versions</i>)